Graded Fault Response for Autonomous Driving via Continuous Sensor Health Monitoring and Risk/Capability-Aware MRM Supervision

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Abstract— Sensor-fault handling in automated driving stacks largely reduces to an immediate in-lane stop, while undetected degradation lets the vehicle keep driving with impaired perception. We propose a graded fault-response layer added onto an unmodified Autoware stack: (i) a behavior-tree monitor estimating per-sensor health as continuous scores (freshness, quality, physics) composed into capability scores, and (ii) an MRM supervisor selecting between shoulder/parking evacuation and deferral to the stock stop by fault type, severity, and remaining capability, with a conservative 3.0 m/s cap on evacuation movement. Across 1,862 valid closed-loop CARLA cases, detection reached 96–100% in every condition except one tolerated by design, with zero false activations, and degraded-state exposure fell from 238–275 m to 11–58 m. Verdicts rest on a dual scheme—an ego collision ledger and a pedestrian physics oracle—built after finding silent collision sensors. A pre-registered dose-response experiment (3 caps × 4 distances × 10 seeds) shows the cap causally moves the avoid/collide boundary inward (8 m: discordant pairs 10:0, exact McNemar $p = 0.002$); encounters in the post-detection deceleration transient remain at risk.

I. INTRODUCTION

Automated driving systems at SAE Level 4 and above cannot presuppose driver intervention: upon a fault they must execute a minimal risk maneuver (MRM) that reaches a minimal risk condition (MRC) on their own. ISO 26262 requires detection and response within the fault-tolerant time interval (FTTI), and SOTIF (ISO 21448) requires managing hazards from sensor performance limitations [1], [2], [3]. In the open-source stack Autoware, the default MRMs are in-lane stops (comfortable/emergency), and its stock diagnostics primarily target communication- and hardware-level anomalies such as topic silence and timeouts [7]. In the version and execution configuration evaluated in this study (Autoware 0.39.0), degradation faults whose values stay within valid ranges left driving to continue without intervention, and no supervisory layer existed that evaluates fault type and degree together with remaining maneuvering capability, selects between evacuation and stopping, and limits evacuation speed to match the fault state. That supervisory layer—not the shoulder maneuver itself—is the contribution of this work; the observation is version-specific, not a claim about Autoware in general.

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An in-lane stop is simple and immediate, but the stopped vehicle blocks following traffic and creates secondary-collision exposure until the lane is cleared. Conversely, driving on with an undetected fault extends the distance exposed to hazards under degraded perception—in our unmonitored observations (one run per fault type) the stock stack completed 238–275 m of route with faults injected. The design space of fault response therefore has three axes: whether to stop now, how far to move, and—if moving—how fast.

Automating human-driver-like shoulder evacuation requires (i) distinguishing faults mild enough to permit moving from faults requiring an immediate stop; (ii) a continuous estimate of fault type and degree (a binary normal/faulty state is insufficient); and (iii) managing the risk of moving itself in a degraded state—the stopping distance against a dart-out. We implement these as a continuous health score, risk/capability-based state transitions, and a conservative speed cap.

Finally, simulation-based safety claims stand only on the reliability of the collision verdict itself: we empirically uncovered a simulator collision-sensor configuration that remained silent, and built a dual adjudication scheme—a real-time ego collision-sensor ledger and a pedestrian physics oracle—on which every safety conclusion rests.

Contributions. (1) A behavior-tree-based monitor that scores per-sensor freshness, quality, and physical consistency independently and composes them over eight-direction spatial sectors into perception/localization capability scores, published as continuous health. (2) A risk/capability-aware MRM supervisor that selects between shoulder/parking evacuation (MRM_MOVE) and deferral to the stock stop (MRM_DEFER) by health-score bands, chooses the refuge by a cost function over degraded-state exposure time, traffic-light observability, and sector deficits, and imposes a fixed 3.0 m/s cap (below the model-derived bound ≈3.89 m/s from $d = v\tau + v^2/2a$) on evacuation movement. (3) A dual collision-adjudication methodology (ego RAW-HIT ledger + pedestrian physics oracle) with a catalogue of verdict pitfalls in runner exit labels and official infraction records. (4) Validation totalling 1,862 valid closed-loop cases, including a pre-registered dose-response experiment verifying the causal cap effect on settled encounters, and transient-vs-settled observations locating residual evacuation risk in the post-detection transient.

The novelty is not in the individual techniques (CUSUM, residual monitoring, noisy-OR fusion,

stopping-distance model—each known), but in the integrated supervisory layer spanning detection through selection and speed, and in the closed-loop demonstration whose verdict reliability is secured at the methodological level.

II. RELATED WORK

Standards. Functional-safety and SOTIF frameworks [1], [2], [3] require runtime monitoring and responses to degradation. ISO 23793-1 [4] gives an MRM type taxonomy and situation-dependent selection principle but covers only in-lane stops (shoulder stops: Part 2, in progress); industrial ISO 3691-4 [5] prescribes "limited speed for limited capability," which our low-speed MRM movement transfers to on-road MRM. None provides a method to continuously estimate fault kind and degree and select response and speed accordingly.

Fault detection and degradation models. Classical FDI—residual generation/decision [6] and sequential change detection (our LiDAR channel borrows CUSUM)—ends at detection; maneuver selection is out of scope. Our fault models follow measured LiDAR soiling [8] and rain scattering [9], and GNSS multipath-type correlated bias [10].

Fallback and MRM planning. Fail-safe trajectory planning [11] maintains stop trajectories and online verification but takes the fault assessment and response choice as given. Emzivat et al. [12] proposed a fallback for impaired monitoring capability; Reschka and Maurer [13] noted that an in-lane stop is not always a safe state—the concern we start from. Closest to us, Xue et al. [14], [15] presuppose a forward-sensor fault and plan safe-zone stops under MPC constraints, and Tang et al. [16] adaptively select MRM behaviors under multiple sensor faults via HAZOP risk analysis with TTC/THW evaluation. They differ in that (i) the fault state is presupposed rather than estimated online; (ii) selection uses surrounding-traffic TTC/THW, whereas we combine risk grace time with remaining perception/localization capability and limit the evacuation speed itself; (iii) validation checks representative scenarios, whereas we verify 1,862 cases on a real stack with methodologically secured encounter verdicts. Skill/ability graphs [17] model capabilities for online decisions; our sector composition couples capability scores directly to MRM selection and the cap. Behavior trees [18] are the implementation structure (parallel composition, execution tracing).

Simulation-based evaluation. CARLA [19] hosts fault-injection studies that inject faults to find vulnerabilities—the opposite direction from ours, where the stack itself detects, grades, and responds. Our infrastructure builds on AWCL [20], the TUM Autoware–CARLA leaderboard coupling. We found no prior report addressing the reliability of simulated collision verdicts at the methodology level.

III. PROPOSED SYSTEM

A. Architecture

The system consists of two independent add-on nodes (Fig. 1). The monitoring module inspects each sensor at 10 Hz on a behavior-tree basis, computing a per-sensor health score in $[0, 1]$ from three detector families: timeout (freshness), quality (noise, valid ratio), and physics (residual, bias). The MRM supervisor subscribes to the scores and runs a state machine: for evacuation it changes the route via the Autoware AD API and imposes a speed cap on the planning layer; for deferral it triggers the stock mrm_handler stop through the diagnostic graph. Stopping-maneuver execution is delegated to the stock stack (goal_planner, mrm_handler); the main stack is not modified. Reception accumulates at native sensor rates; evaluation runs at 10 Hz, consistent with AUTOSAR watchdog-style periodic supervision.

B. Sensor Scoring

Freshness. For elapsed time $\mathrm{age}(t)$ since last reception, $s_{\mathrm{fresh}} = \max(0,\, 1 - \mathrm{age}/\tau_{\mathrm{detect}})$ decays linearly—so the score can cross the downstream band boundary before the budget is exhausted, enabling early transition under complete silence. A k-miss rule penalizes early when nothing arrives for $3\times$ the moving median of the reception interval.

Quality. For LiDAR, a Gaussian noise level $\hat{\sigma} = \overline{|\Delta r|} \cdot \sqrt{\pi}/2$ (neighboring-point range differences, $|\Delta r| < 1$ m) is referenced to a boot-time baseline that refuses to track abrupt rises (so the fault cannot become its own baseline); the score is $\exp(-\max(0, \hat{\sigma} - \mathrm{base})/0.1)$ times a valid-point-ratio score, with a parallel CUSUM test so detection delay is inversely proportional to fault magnitude. Camera quality applies the same structure on inter-frame pixel differences (no cumulative test: pixel difference rises with scene motion). Sparsification is caught by the valid-ratio channel, clutter by the range-discontinuity channel.

Localization (three channels). (a) a residual channel on the GNSS–EKF distance (sensitive to noise and multipath-type correlated bias); (b) a bias channel comparing 3-s means of IMU and EKF yaw rates (normal turning cancels; gyro bias remains on the IMU side); (c) a silence channel that decays the score when GNSS input stops $\geq$1 s while driving—covering the residual channel's blind spot of freezing at the last value.

C. Sector and Capability Synthesis

Sensor scores map to eight directional sectors by mounting direction and field of view and fuse as $s_{\mathrm{sector}} = 1 - \prod_i (1 - s_i)$; perception capability is a sector-weighted sum (front-weighted), combined with traffic-light observability at 0.6:0.4; the system health is $\min(\mathrm{perception}, \mathrm{cap}_{\mathrm{loc}})$—the weakest-link principle.

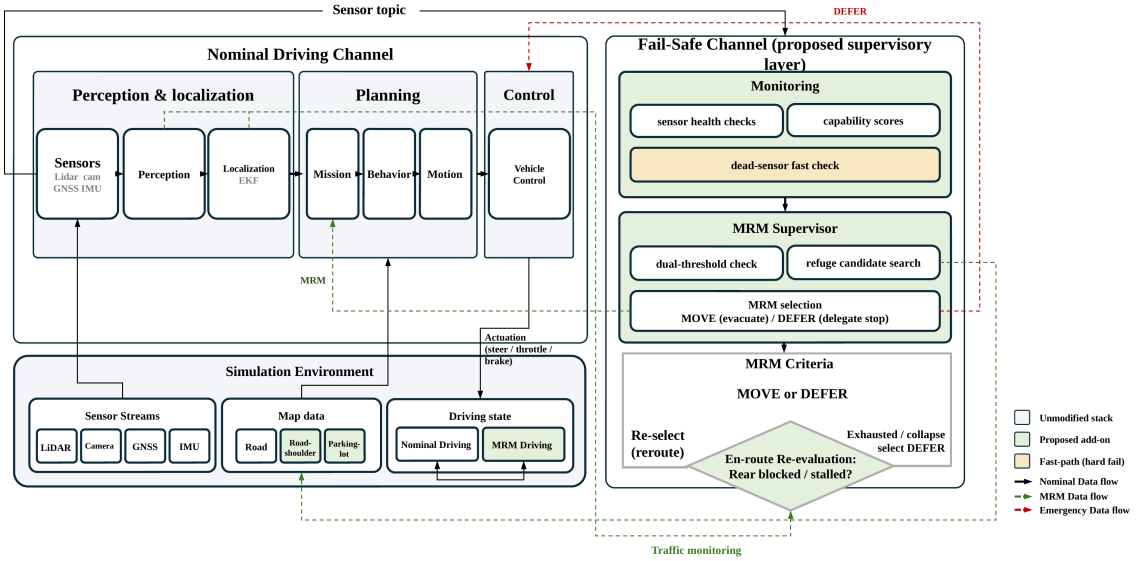


Fig. 1. Dual-channel structure. Left: the nominal driving channel of unmodified Autoware; right: the proposed supervisory layer—monitoring subscribes read-only to sensor topics and produces health/capability scores; the decision stage selects MOVE (evacuate: reroute with a speed cap) or DEFER (inject a diagnostic ERROR to trigger the stock stop), re-evaluating rear occupancy and stalls en route. Bottom: the CARLA closed loop; refuge search draws on annotated map data.

D. Risk-Coupled Detection Budget

The freshness decay constant is coupled to driving risk: $\tau_{\text{detect}}(t) = \min(\tau_{\text{fresh}}, \text{TTC}_{\text{cons}}(t))$, where TTC_{cons} is the minimum of nearest-obstacle TTC and path-crossing time with a fail-conservative hold when the TTC source itself is unhealthy. The design intent is $T_{\text{detect}} + T_{\text{response}} + T_{\text{margin}} \leq T_{\text{hazard}}$; the implementation is a risk-adaptive heuristic approximation of this budget, not a formal guarantee (explicit FRTI/margin reservation is future work). For degradation faults, the requirement is a severity curve $T_{\text{detect}}(\sigma) \leq T_{\text{hazard}}(\sigma)$: small faults are caught later but become hazardous later.

E. MRM Supervisor

Dual thresholds on health (upper 0.7, lower 0.3) span three bands: NORMAL, evacuation (MRM_MOVE), and deferral (MRM_DEFER, on floor entry or localization-confidence collapse). Decay constants were designed so a wide insensitive gap lies between faulted and normal scores; behavior follows the band structure, not precise threshold placement. Timing splits by evidence type: degradation faults are statistical evidence requiring sustained confirmation (1.0 s band entry; 2.0 s at the floor since deferral is terminal; a 5.0 s recovery grace, unchanged under a 0.5–5.0 s sensitivity sweep), whereas hard fails (silence past the 850 ms freshness deadline) already carry accumulated evidence: confirmations are skipped and unrecoverable conditions defer immediately. Selection is not frozen while moving: sustained rear occupancy demotes reversing parking candidates to a shoulder stop, stalls get bounded retries, and candidate exhaustion or en-route health collapse falls back to DEFER.

Refuge selection minimizes

$$J(c) = w_t\,\eta(c) + w_{tl}\,n_{tl}(c)\,g(tl) + w_{sec}\sum(1 - s_{\text{req}}) + w_{loc}\,\eta(c)(1 - \text{loc}), \quad (1)$$

where $\eta(c)$ is degraded-state exposure time (route time + expected signal waits)—risk accumulates per unit time, not distance. Route times and traffic-light counts are computed on an in-house road-network graph parsed directly from the lanelet2 map, keeping the decision layer outside the influence of main-stack faults. The key safeguard is the traffic-light observability gate: a candidate whose route passes signals no healthy camera can read is excluded with $J = \infty$. Shoulder/parking candidates are manually annotated map attributes (16 shoulders, 4 lots on the test map).

F. Conservative Low-Speed MRM Movement (Speed Cap)

Evacuation is movement under degraded perception, so stopping capability against a dart-out dominates. With reaction time τ and deceleration a, stopping from speed v requires

$$d_{\text{safe}}(v) = v\tau + \frac{v^2}{2a} + d_{\text{margin}} \leq D_{\text{avail}}. \quad (2)$$

Here $\tau \approx 1.5$ s, $a \approx 3.5$ m/s^2 are nominal lumped values calibrated to the measured ladder boundary—a design approximation for choosing the cap policy value, not a predictive safety model. On entering MRM_MOVE the supervisor imposes a fixed 3.0 m/s cap, chosen inside the band whose upper end is the model-inverted cap that still stops within an 8 m dart-out (≈3.9 m/s) and whose lower end bounds evacuation-completion time; no optimization within the band is claimed. The cap is

delivered to the input of Autoware's velocity-limit arbiter and its delivery is verified by measured vehicle speed, not publish records—an interface mismatch that silently dropped the cap in earlier campaign runs was discovered exactly this way and corrected (Sec. V-D).

IV. EXPERIMENTAL SETUP

A. Environment and Campaign Matrix

The closed loop couples CARLA 0.9.16 with Autoware 0.39.0 over the AWCL bridge [20] on Town07 (custom vector map with annotated refuges; speed limit 15 km/h); the simulator is fully restarted for every case. A case is {method, fault, severity, traffic, repetition, injection time}; methods are ours (proposed supervisor), baseline (identical detection, deferral to the stock stop—isolating response strategy from detection), and mrm-off (passive scoring only). The main campaign is fault $\times$ severity $\times$ traffic density (none/defensive/aggressive) $\times$ method with 1,131 valid cases (39 short of the 1,170 plan, all in high-severity LiDAR clutter); a navigation-extension matrix with identical structure adds four navigation faults (611 valid), and encounter experiments complete the 1,862 total. The fault model injects inside the agent's sensor pipeline: camera death/noise (cf/cn), LiDAR death/sparsification/clutter (lf/lso/lcl, physical models [8], [9]), GNSS noise/correlated bias/outage (gn/gb/gf [10]), and IMU noise/bias/drift (imn/imb/imd), with severities low/mid/high where applicable; injection times and fault waveforms are seed-fixed for pairing.

B. Collision Adjudication

The initial implementation adjudicated collisions with a collision sensor attached to the pedestrian; measurement showed this sensor never fired in CARLA, and a listener death additionally lost separation records, so non-intervention-arm cases were being misjudged as avoided (all affected data were discarded and re-run). The reconstructed system has two layers: (i) primary: a real-time ledger (RAW-HIT) written by the patched callback of the ego's sensor.other.collision the moment a collision occurs, with counterpart actor identifiers (endpoint-structure grazes are classified separately)—unlike official infraction tallies it survives early/forced termination; (ii) audit: a pedestrian physics oracle (kinematic anomalies: flung >4.5 m/s or a >5 m jump within 1 s) and the per-tick separation stream. Runner exit labels are refused as evidence (the collision label can include endpoint contacts, the hazardous-stop label post-collision stops, and official infractions survive only natural termination). All encounter-level figures here are ledger-based; latencies use simulation time (actor destruction stalls the simulator $\sim$1.6 s).

C. Encounter Experiments

A decoupled injector process spawns pedestrian dart-out crossings at the right lane edge with per-seed fixed walking speed, lateral jitter, and fault waveform. Designs: (i) a fault-free distance ladder at 4.2 m/s, spawn distances $\{14, 11, 8, 6, 5\}$ m, n=4 each; (ii) the same ladder blinded (lso-high); (iii) an 8 m encounter comparison under gn-mid (EKF input contaminated, perception intact—so all arms share detection capability); (iv) a low-speed ladder at 3.0 m/s cruise; and (v) a dose-response experiment: caps $\{4.2, 3.6, 3.0\}$ m/s $\times$ distances $\{6, 7, 8, 9\}$ m $\times$ 10 seeds = 120 runs, launched by a settlement trigger—the pedestrian spawns D m ahead only when the supervisor is in MRM_MOVE and measured speed has stayed in the cap-specific band for 1.0 s—so an encounter under an actually holding cap is guaranteed at trigger level. The primary endpoint (8 m, paired 4.2 vs 3.0 exact McNemar) and a manipulation-check adoption rule (first-tick speed in band $\pm$0.15 m/s, ahead-of-ego geometry, launch within 420 s) were pre-specified; sample size was fixed at 120 adopted runs.

V. RESULTS

A. Detection Performance

Table I summarizes the campaign: detection rates of 96–100% in 19 of 20 fault conditions—median latencies $\sim$1.0 s for hard fails (ledger, simulation time) and 2.1–8.6 s for degradation faults (wall clock; different clock domains)—with zero false activations in 45 fault-free controls. The sole exception, mid-severity LiDAR sparsification (2%), lies by design in the tolerated band ($\sim$55 m effective range remains; the supervisor deliberately continues driving); since tolerance cannot yet be separated from channel insensitivity, we read the table as intervention eligibility rather than fault-existence detection. Notable structure: IMU-drift latency is inversely related to ramp slope (8.6 vs 5.4 s)—small faults are caught later but become dangerous later; GNSS-outage latency is bimodal (an intended delay: a vehicle stopped at a red light is not a risk exposure); and deferral stops split by family (GNSS $\rightarrow$ comfortable, IMU $\rightarrow$ hard stops—corrupted rotational sense demotes trajectory following).

Behind the binary rates lies the continuous signal: the decision variable $\min(\text{perception}, \text{cap}_{\text{loc}})$ separates into distinct bands by fault intensity—fault-free runs never fell below 0.71 after warm-up (consistent with zero false activations), mid severities stay in the tolerated band, and high severities depart immediately on injection—the basis of the graded normal/evacuate/defer structure.

B. Graded Evacuation Behavior and Exposure

For faults preserving the capability to move, the proposal evacuated while the baseline (by definition) stopped in lane: camera death 16/45 shoulder evacuations (27 intended deferrals, 2 maneuver incompletions), camera noise 25/90, GNSS noise 42/90. Severity determined behavior most sharply in the navigation family: gn-mid produced 40/45 shoulder evacuations,

TABLE I

Detection rate and median latency per fault condition (n=45 unless noted; †: navigation extension; *: simulation time, others wall clock).

Fault	Sev.	Det. rate	Median lat. (s)
cf camera death	–	1.00	0.98*
cn camera noise	mid/high	0.98/1.00	3.3/3.2
lf LiDAR death	–	1.00	1.00*
lso LiDAR sparsif.	mid/high	0.02/0.98	– (tolerated band)/5.0
lcl LiDAR clutter	mid/high[a]	1.00/0.96	4.8/4.5
gn GNSS noise	mid/high	1.00/1.00	2.1/4.4
imn IMU noise	mid/high	1.00/0.98	4.8/3.7
gb† GNSS corr. bias	mid/high	1.00/1.00	3.0/2.3
gf† GNSS outage	–	1.00	3.2 (driving)[b]
imb† IMU gyro bias	mid/high	1.00/1.00	3.6/3.2
imd† IMU bias ramp	mid/high	1.00/1.00	8.6/5.4
nf fault-free control	–	0 false activations / 45	

[a] lcl-high n=26 (final batch halted; accounted).
[b] bimodal: 21–40 s when the outage coincides with a signal stop.

whereas gn-high converged to deferral stops in 42/45—movement is abandoned when localization confidence falls below the movement minimum. Conversely, lso-mid continued driving deliberately (response proportional to degree, not stop-at-every-anomaly). Unmonitored (mrm-off, one run per type) the stack completed 238–275 m in the faulted state; with monitoring, exposure became a policy variable: 32–58 m for detect+evacuate, 9–18 m for detect+stop (11–12 m for LiDAR clutter). The stop minimizes exposure at the cost of lane occupancy; evacuation accepts tens of meters of controlled exposure for lane clearance and an off-lane refuge.

C. Avoid/Collide Boundary and the 8 m Encounter

The fault-free ladder at 4.2 m/s produced a clean staircase: 14 and 11 m avoided 4/4 each; 8, 6, 5 m collided 4/4 each—boundary in 8–11 m, matching Eq. (2)'s 8.8 m (Fig. 2; the same data calibrated the model—self-consistency, not independent validation). Blinded (lso-high), the stock stack froze mid-road before the encounter at 14/11 m and collided at $\leq$8 m: an unmonitored degraded vehicle either stands still or strikes—neither an MRC.

Under gn-mid (perception intact, so all arms share detection capability), an 8 m dart-out gave: stock at 4.2 m/s collided 10/10; the proposed low-speed settled cruise avoided 8/8 and completed evacuation (auxiliary Fisher $p = 2.3 \times 10^{-5}$); and—critically—the same policy collided 2/2 when the encounter fell in the deceleration transient (3.5–3.8 m/s). Protection begins only after deceleration completes. The low-speed ladder placed the 3.0 m/s boundary at 7–8 m (8 m: 8/8 avoided; 7/6/5 m: 2/2 collisions each).

D. Dose-Response Verification of the Speed Cap

Because the two ladders used different launch protocols, the cap effect was verified in a controlled pre-registered experiment on settled encounters only, after fixing a cap-delivery interface defect (the earlier "capped" runs were receiving no cap—publish records are

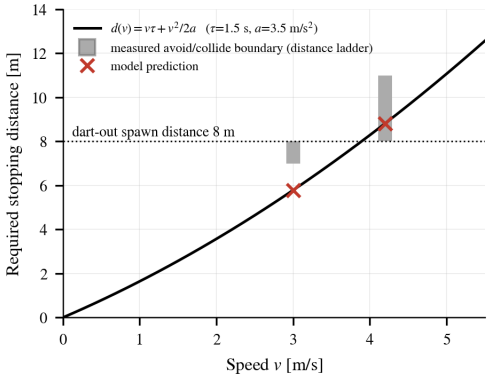


Fig. 2. Stopping-distance model $d = v\tau + v^2/2a$ (τ=1.5 s, a=3.5 m/s^2) vs. measured boundaries (shaded). At 4.2 m/s the measured 8–11 m matches the model's 8.8 m; at 3.0 m/s the model's 5.79 m is a lower bound of the measured 7–8 m. Dashed: the 8 m spawn.

TABLE II

Dose-response: collisions per cell (n=10 seeds, ledger adjudication, settled encounters).

Cap (m/s)	6 m	7 m	8 m	9 m	Boundary
4.2 (control)	10/10	10/10	10/10	10/10	>9 m
3.6	10/10	10/10	10/10	2/10	∼9 m
3.0 (policy)	8/10	5/10	0/10	0/10	∼7 m

not delivery; measured speed is). All 120 adopted runs passed the manipulation check (per-arm first-tick speeds 4.05–4.13, 3.37–3.58, 2.60–3.07 m/s; five failing runs re-run blind to outcome). Results (Table II, Fig. 3) are monotone along both axes. At the pre-specified primary endpoint (8 m), cap 4.2 collided in all ten seeds and cap 3.0 in none—discordant pairs 10:0, exact McNemar two-sided $p = 0.002$: once the cap actually holds, its collision-reduction effect is detected with unanimous seed-level directionality. The boundary moves inward monotonically: beyond 9 m (4.2), ∼9 m (3.6, 2/10), ∼7 m (3.0, 5/10 at 7 m, 0/10 at $\geq$8 m), mutually corroborating the low-speed ladder's 7–8 m interval. Interpretation limits: the causal conclusion is restricted to the settled regime (the cap creates the post-settling margin, not transient protection); no optimality is claimed for 3.0 m/s; cells are n=10, single fault, single corridor.

E. Hard-Fail Decision Latency

Hard fails are decided by an evidence-aware fast path (skipping noise-oriented debounce and futile recovery holds) against an internal 1.5 s detection-to-decision design deadline (not an FTTI-compliance claim). Over the full camera/LiDAR-death matrices, ledger-based simulation-time latencies gave medians ≈1.0 s, P95 1.05–1.15 s, maximum 1.20 s, zero violations, with MRM attainment maintained in full.

VI. DISCUSSION AND LIMITATIONS

Stop vs. evacuation is a temporal trade-off. The in-lane stop showed the lowest collision rate even under

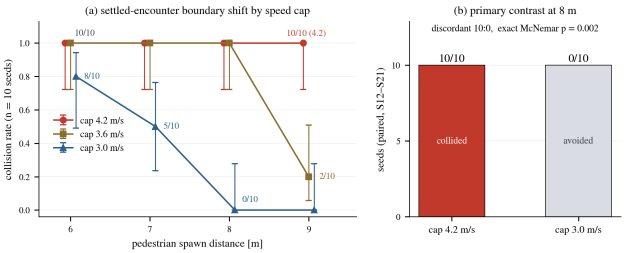


Fig. 3. Dose-response experiment (120 runs, ledger adjudication). (a) Collision rate vs. spawn distance per cap (Wilson 95% CI). (b) Primary contrast at 8 m: 10/10 vs. 0/10, discordant 10:0, exact McNemar $p = 0.002$.

transient-window launches, at the price of sustained lane occupancy; evacuation accepts controlled moving exposure and residual transient risk but on success clears the lane and reaches an off-lane refuge (lane clearance is relinquished occupancy, not proven final-position safety). Neither is unconditionally superior; the choice should follow the fault's temporal structure—concrete selection variables for ISO 23793-1.

Residual risk is temporally localized. All evacuation-arm collisions occurred before deceleration completed (transient 2/2), none after settling (0/8). Distinguishing the post-detection transient from the overall fault-response risk window ($T_{\text{detect}} + T_{\text{decel}}$), the results support—observationally, not as a causal manipulation of detection timing—that earlier detection and faster deceleration shrink this window, widening the margin within the FTTI.

Adjudication lessons generalize. "Zero collisions" is meaningful only after the verdict channel is proven alive: positive-control checks of collision sensors, physics-oracle cross-checks, and refusal to read absence of a record as absence of an event. The silently undelivered cap carries the same lesson—actuation must be verified by downstream effect. Both failure modes arise structurally in automated large-scale experiments; the real-time ledger was the remedy.

Limitations. Single map, single 15 km/h speed limit, two launch corridors: the numeric boundaries and dose-response values are low-speed-ODD numbers; generalizable claims are structural (the stopping-distance dependence of the boundary, transient risk concentration, the stop/evacuation trade-off). Autoware observations are version-specific (0.39.0). Dose-response cells are n=10 with wide CIs; testing was restricted to the pre-specified endpoint. Campaign-level collision-free status predates the ledger (no pedestrian injections there; stated as a consistency gap). The dynamic detection budget was not ablated against a fixed deadline; hazard-time-referenced compliance requires an instrumentation extension; the health-proportional cap is implemented but unevaluated; no real-vehicle validation.

VII. CONCLUSION

We proposed and validated, in an Autoware–CARLA closed loop of 1,862 valid cases, a supervisory add-on layer that estimates sensor faults as continuous health scores and selects MRMs in a graded manner, with a conservative speed cap on evacuation movement. Three conclusions carry beyond the numbers. First, shoulder evacuation is no panacea: stopping and evacuation form a temporal trade-off between immediate collision risk and lane clearance, and the choice should follow the fault's temporal structure. Second, evacuation's residual risk concentrates in the post-detection transient, while settled low speed verifiably moves the avoidance boundary inward (transient 2/2 vs. settled 0/8; dose-response 10:0, $p = 0.002$)—supporting earlier detection and faster deceleration within the FTTI. Third, simulation-based safety claims hold only on a validated adjudication framework; the ledger/physics-oracle scheme and catalogued verdict pitfalls are reusable for any simulated MRM comparison. Future work: high-speed ODDs, multiple maps, formalizing the detection budget, the health-proportional cap, and real-vehicle validation.

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